Simple inelastic collisions

(P) Nucleus of (ground state) mass m is in excited state with energy $\Delta\mathcal{E}$. It emits photon (γ particle) and goes into ground state. Find photon's freq. ω.

$$p_0^i = \left(\frac{mc^2+\Delta\mathcal{E}}{c}, 0\right) \qquad p_N^i = \left(\frac{\mathcal{E}_N}{c}, \vec{p}_N\right)$$

$$p_\gamma^i = \left(\frac{\hbar\omega}{c}, \vec{p}_\gamma\right)$$

$$\begin{cases} mc^2 + \Delta\mathcal{E} = \hbar\omega + \mathcal{E}_N \\ 0 = \vec{p}_N + \vec{p}_\gamma \end{cases}$$

But $p_\gamma = \frac{\hbar\omega}{c} \Rightarrow p_N = \frac{\hbar\omega}{c}$

$$\left(\frac{\mathcal{E}_N}{c}\right)^2 - p_N^2 = m^2c^2 \Rightarrow \mathcal{E}_N^2 = \sqrt{m^2c^4 + c^2p_N^2} =$$

$$= \sqrt{m^2c^4 + (\hbar\omega)^2}$$

$$\Rightarrow mc^2 + \Delta\mathcal{E} = \hbar\omega + \sqrt{(\hbar\omega)^2 + m^2c^4}$$

$$(mc^2 + \Delta\mathcal{E} - \hbar\omega)^2 = (\hbar\omega)^2 + m^2c^4$$

$$\cancel{m^2c^4} + \Delta\mathcal{E}^2 + \cancel{(\hbar\omega)^2} - 2\hbar\omega(mc^2+\Delta\mathcal{E}) + 2mc^2\Delta\mathcal{E} =$$

$$= \cancel{(\hbar\omega)^2} + \cancel{m^2c^4}$$

$$2\hbar\omega = \frac{\Delta\mathcal{E}^2 + 2mc^2\Delta\mathcal{E}}{mc^2+\Delta\mathcal{E}} = \Delta\mathcal{E}\,\frac{(\Delta\mathcal{E}+2mc^2)}{\Delta\mathcal{E}+mc^2}$$

$$\hbar\omega = \frac{\Delta\mathcal{E}}{2}\,\frac{\Delta\mathcal{E}+2mc^2}{\Delta\mathcal{E}+mc^2} = \frac{\Delta\mathcal{E}}{2}\,\frac{2mc^2\left(1+\frac{\Delta\mathcal{E}}{2mc^2}\right)}{mc^2\left(1+\frac{\Delta\mathcal{E}}{mc^2}\right)} = \Delta\mathcal{E}\left(1 - \frac{\Delta\mathcal{E}}{2mc^2}\right)$$

But $\Delta\mathcal{E} + mc^2 = Mc^2$ where M = mass of the atom in excited state

$$mc^2 = Mc^2 - \Delta\mathcal{E} = Mc^2\left(1 - \frac{\Delta\mathcal{E}}{Mc^2}\right)$$

$$\hbar\omega = \Delta\mathcal{E}\left(1 - \frac{\Delta\mathcal{E}}{2mc^2}\right) = \Delta\mathcal{E}\left(1 - \frac{\Delta\mathcal{E}}{2Mc^2\left(1 - \frac{\Delta\mathcal{E}}{Mc^2}\right)}\right) =$$

$$= \Delta\mathcal{E}\left(1 - \frac{\Delta\mathcal{E}}{2Mc^2} + o\left(\frac{\Delta\mathcal{E}}{Mc^2}\right)^2\right)$$

$\Rightarrow \hbar\omega < \Delta\mathcal{E}$ due to atom's recoil.

In the crystal, the atom is firmly implanted in the lattice, and its recoil is suppressed. (Moessbauer effect)

<u>Compton scattering</u> $e^- + \gamma \to e^- + \gamma$

before
$$\begin{cases} \rightsquigarrow & p_1^\alpha = \left(\frac{\varepsilon_0}{c}, \hbar k, 0, 0\right) \\ e^- & p_2^\alpha = (m_2 c, 0, 0, 0) \end{cases}$$

$\varepsilon_0 = \hbar\omega$

$\hbar k = \frac{\hbar\omega}{c} = \frac{\varepsilon_0}{c}$

after
$$\begin{cases} \rightsquigarrow & q_1^\alpha = \left(\frac{\varepsilon_1}{c}, \hbar k_1 \cos\theta, \hbar k_1 \sin\theta, 0\right) \\ & q_2^\alpha = (m_2 c, q_2 \cos\theta_2, q_2 \sin\theta_2, 0) \end{cases}$$

$\varepsilon_1 = \hbar\omega_1$

$$p_1^\alpha + p_2^\alpha = q_1^\alpha + q_2^\alpha$$

$$p_2^\alpha = q_1^\alpha + q_2^\alpha - p_1^\alpha$$

$$p_2^\alpha p_{2\alpha} = m_2^2 c^2 = \text{invariant}$$

Similarly, after the collision

$$q_2^\alpha = p_1^\alpha + p_2^\alpha - q_1^\alpha$$

But $q_2^\alpha q_{2\alpha} = m_2^2 c^2 = (p_1 + p_2 - q_1)^\alpha (p_1 + p_2 - q_1)_\alpha$

$$= \cancel{p_1^\alpha p_{1\alpha}} + \underline{p_1^\alpha p_{2\alpha}} - p_1^\alpha q_{1\alpha}$$
$$+ \underline{p_2^\alpha p_{1\alpha}} + \underbrace{p_2^\alpha p_{2\alpha}}_{m_2^2 c^2} - p_2^\alpha q_{1\alpha}$$
$$- q_1^\alpha p_{1\alpha} - q_1^\alpha p_{2\alpha} + \cancel{q_1^\alpha q_{1\alpha}}$$

$p_1^\alpha p_{1\alpha} = q_1^\alpha q_{1\alpha} = 0$ since photon is massless.

$\Rightarrow$ we obtain $2\left(p_1^\alpha p_{2\alpha} - p_1^\alpha q_{1\alpha} - p_2^\alpha q_{1\alpha}\right) = 0$

$$p_1^\alpha p_{2\alpha} = m_2 \mathcal{E}_0$$

$$p_1^\alpha q_{1\alpha} = \frac{\mathcal{E}_0 \mathcal{E}_1}{c^2} - \hbar k \cdot \hbar k_1 \cos\theta = \frac{\mathcal{E}_0 \mathcal{E}_1}{c^2}(1-\cos\theta)$$

$$p_2^\alpha q_{1\alpha} = m_2 \mathcal{E}_1$$

$$\Rightarrow \quad m_2 \mathcal{E}_0 - m_2 \mathcal{E}_1 = \frac{\mathcal{E}_0 \mathcal{E}_1 (1-\cos\theta)}{c^2}$$

$$\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} = \frac{1-\cos\theta}{m_2 c^2}$$

Since $\mathcal{E} = \hbar\omega = \frac{hc}{\lambda} \Rightarrow \frac{1}{\mathcal{E}_1} = \frac{\lambda_1}{hc}$ $\quad (h = 2\pi\hbar)$

$$\Rightarrow \lambda_1 - \lambda_0 = \frac{h}{m_2 c}(1-\cos\theta) \quad \text{Compton formula}$$

$$\frac{h}{m_2 c} = \frac{h}{m_e c} = \text{Compton wavelength}$$

$$= 2.426 \cdot 10^{-12}\,\text{m}.$$

Vavilov - Cherenkov effect

Charged particle is uniformly moving in dielectric medium with refraction index $n(\omega)$.
Mass m, charge e, speed v_0.
Show that it can emit EM waves (photons) and find angle θ between photon's momentum $\hbar\vec{k}$ and $\vec{v}_0$.

before $\quad p_0^i = p^i + k^i \quad$ after

$$\left(\frac{\mathcal{E}_0}{c}, \vec{p}_0\right) = \left(\frac{\mathcal{E}}{c}, \vec{p}\right) + \left(\frac{\hbar\omega}{c}, \hbar\vec{k}\right)$$

$$k = |\vec{k}| = \frac{\omega}{c/n} = \frac{n\omega}{c}$$

$$\Rightarrow \quad p_0^i - k^i = p^i$$

$$\begin{cases} (\mathcal{E}_0 - \hbar\omega)^2 = \mathcal{E}^2 \\ (\vec{p}_0 - \hbar\vec{k})^2 = \vec{p}^2 \end{cases} \Rightarrow \quad p_0^2 + (\hbar\vec{k})^2 - 2\hbar p_0 k \cos\theta = p^2 \quad (1)$$

$$\mathcal{E} = \sqrt{m^2c^4 + c^2\vec{p}^2} \qquad \vec{p} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}} \Rightarrow \vec{p} = \frac{\mathcal{E}}{c^2}\vec{v}.$$

$$\mathcal{E} = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

$$\mathcal{E}_0^2 + (\hbar\omega)^2 - 2\hbar\omega\mathcal{E}_0 = \mathcal{E}^2$$

$$m^2c^4 + c^2p_0^2 + (\hbar\omega)^2 - 2\hbar\omega\mathcal{E}_0 = m^2c^4 + c^2p^2$$

$$c^2(p_0^2 - p^2) = 2\hbar\omega\mathcal{E}_0 - \hbar^2\omega^2. \quad (2)$$

From (1) (2) : $\cos\theta = \dfrac{p_0^2 - p^2 + \hbar^2 k^2}{2\hbar p_0 k} = \dfrac{\hbar k}{2p_0} +$

$$+ \frac{2\hbar\omega\varepsilon_0 - \hbar^2\omega^2}{c^2 \cdot 2\hbar k p_0} = \frac{\hbar k}{2p_0} + \frac{\omega\varepsilon_0}{k c^2 p_0} - \frac{\hbar\omega^2}{2kp_0c^2}$$

$$= \frac{\hbar k}{2p_0} + \frac{\omega\varepsilon_0}{\frac{n\omega}{c} c^2 p_0} - \frac{\hbar\omega^2}{2c^2 p_0 \frac{n\omega}{c}} =$$

$$= \frac{\hbar k}{2p_0} - \frac{\hbar\omega}{2nc p_0} + \frac{\varepsilon_0}{c p_0 n} =$$

$$= \frac{\hbar}{2p_0}\frac{n\omega}{c} - \frac{\hbar\omega}{2nc p_0} + \frac{\varepsilon_0}{c p_0 n} =$$

$$= \frac{\hbar\omega}{2nc p_0}(n^2-1) + \underbrace{\left(\frac{\varepsilon_0}{c p_0}\right)}_{\frac{c}{V_0}} \frac{1}{n} = \frac{1}{n V_0/c} + (n^2-1)\frac{\hbar\omega}{2nc p_0}$$

$$\cos\theta = \frac{1}{n V_0/c}\left(1 + (n^2-1)\frac{\hbar\omega}{2c}\frac{V_0}{c p_0}\right)$$

$$\frac{\hbar\omega}{2c} = \frac{\hbar}{2c}\frac{2\pi c}{n\lambda} = \frac{\pi\hbar}{n\lambda}$$

$$\lambda = 2\pi\frac{c/n}{\omega} \Rightarrow \omega = \frac{2\pi c}{n\lambda}$$

$$\frac{V_0}{c p_0} = \frac{c}{\varepsilon_0} = \frac{c}{\frac{mc^2}{\sqrt{1 - V_0^2/c^2}}} = \frac{\sqrt{1 - V_0^2/c^2}}{mc}$$

$$\cos\theta = \frac{c}{n v_0}\left(1 + (n^2-1)\,\frac{\pi\hbar}{n\lambda}\,\frac{\sqrt{1-v_0^2/c^2}}{mc}\right) =$$

$$= \frac{c}{n v_0}\left(1 + \frac{\pi \Lambda}{n\lambda}\,(n^2-1)\sqrt{1-\frac{v_0^2}{c^2}}\right)$$

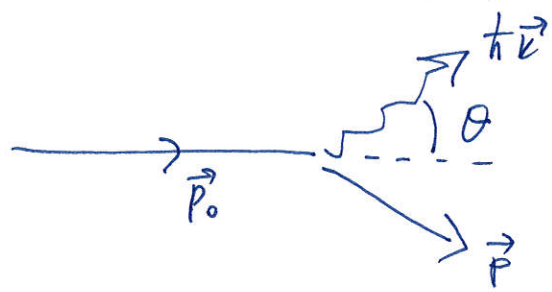
where $\Lambda = \frac{\hbar}{mc}$ = Compton wavelength

Omitting quantum $\left(\sim \frac{\Lambda}{\lambda}\right)$ terms,

$$\cos\theta \approx \frac{c}{n v_0} \leq 1 \Rightarrow v_0 \geq \frac{c}{n}$$

Possible for particle with $\frac{c}{n} < v_0 < c$

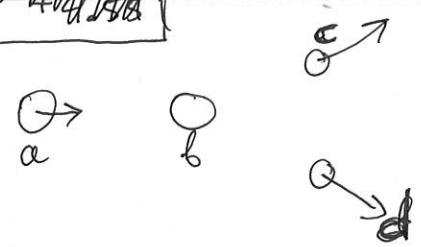
$\Rightarrow$ needs media with $n > 1$.



$\Lambda_{electron} \sim 10^{-12}\,m$

$\lambda_{light} \sim 10^{-6}\,m$

$\Rightarrow \frac{\Lambda}{\lambda} \ll 1$ is a good approx.



$a + b \to c + d$

(Note that in general $m_a \neq m_c$, $m_b \neq m_d$)

$$p^\alpha = \left(\frac{\mathcal{E}}{c}, \vec{p}\right) \Rightarrow p^\alpha p_\alpha = \frac{\mathcal{E}^2}{c^2} - \vec{p}^{\,2} = m^2 c^2$$

Introduce invariants s, t, u

$$s = (p_a + p_b)^\alpha (p_a + p_b)_\alpha \,, \quad t = (p_a - p_c)^\alpha (p_a - p_c)_\alpha$$

$$u = (p_a - p_d)^\alpha (p_a - p_d)_\alpha$$

$$\begin{aligned} s + t + u = {} & m_a^2 c^2 + m_b^2 c^2 + 2 p_a^\alpha p_{b\alpha} \\ & + m_a^2 c^2 + m_c^2 c^2 - 2 p_a^\alpha p_{c\alpha} \\ & + m_a^2 c^2 + m_d^2 c^2 - 2 p_a^\alpha p_{d\alpha} \end{aligned}$$

~~In CM frame: $\vec{p}_a + \vec{p}_b = 0 = \vec{p}_c + \vec{p}_d$~~

$$\begin{aligned} s + t + u = {} & (m_a^2 + m_b^2 + m_c^2 + m_d^2) c^2 \\ & + 2 m_a^2 c^2 + 2 p_a^\alpha \underbrace{(p_{b\alpha} - p_{c\alpha} - p_{d\alpha})}_{-p_{a\alpha}} \end{aligned}$$

But $p_a^\alpha + p_b^\alpha = p_c^\alpha + p_d^\alpha \Rightarrow$

$$2\left(m_a^2 c^2 - \underbrace{p_a^\alpha p_{a\alpha}}_{\frac{\mathcal{E}_a^2}{c^2} - \vec{p}_a^{\,2}}\right) = 2\left(m_a^2 c^2 + \vec{p}_a^{\,2} - \frac{\mathcal{E}_a^2}{c^2}\right) = 0$$

$$\Rightarrow s + t + u = (m_a^2 + m_b^2 + m_c^2 + m_d^2) c^2$$

Let b ~~not move~~ be initially at rest in lab frame $\Rightarrow \mathcal{E}_b = m_b c^2$, $\vec{p}_b = 0$

$$\Rightarrow s = m_a^2 c^2 + m_b^2 c^2 + 2\frac{\mathcal{E}_a}{c}\frac{\mathcal{E}_b}{c} - 2\vec{p}_a \cdot \cancel{\vec{p}_b}^{\,0}$$

$$\Rightarrow \mathcal{E}_a = \frac{s - m_a^2 c^2 - m_b^2 c^2}{2 m_b}$$

In CM frame (K') $\underline{\vec{p}'_a = -\vec{p}'_b}$, $\vec{p}'_c = -\vec{p}'_d$

$$\Rightarrow 2\frac{\mathcal{E}'_a \mathcal{E}'_b}{c^2} - \vec{p}'_a \cdot (-\vec{p}'_a) - (-\vec{p}'_b)\cdot \vec{p}'_b$$

$$= 2\frac{\mathcal{E}'_a \mathcal{E}'_b}{c^2} + \left(\frac{\mathcal{E}'^2_a}{c^2} - m_a^2 c^2\right) + \left(\frac{\mathcal{E}'^2_b}{c^2} - m_b^2 c^2\right)$$

$$\Rightarrow s' = s = \frac{1}{c^2}\left(\mathcal{E}'^2_a + \mathcal{E}'^2_b + 2\mathcal{E}'_a \mathcal{E}'_b\right) =$$

$$= \frac{1}{c^2}\left(\mathcal{E}'_a + \mathcal{E}'_b\right)^2 \Rightarrow s = \frac{1}{c^2}\left(\mathcal{E}'_c + \mathcal{E}'_d\right)^2$$

$\Rightarrow$ the meaning of $\underline{s}$: square of the total energy in CM frame K'.

$$t + u = 2m_a^2 c^2 + m_c^2 c^2 + m_d^2 c^2 - 2\underbrace{p'^{\alpha}_a (p'_c + p'_d)_{\alpha}}_{\frac{\mathcal{E}'_a}{c}\,\frac{\mathcal{E}'_c + \mathcal{E}'_d}{c} - \vec{p}'_a \cdot \underbrace{(\vec{p}'_c + \vec{p}'_d)}_{\emptyset}}$$

$$\Rightarrow t + u = \left(2m_a^2 + m_c^2 + m_b^2\right)c^2 - \frac{2\mathcal{E}'_a(\mathcal{E}'_c + \mathcal{E}'_d)}{c^2}$$

but $\mathcal{E}'_c + \mathcal{E}'_d = c\sqrt{s}$

$$\underbrace{t + u}_{(m_a^2 + m_b^2 + m_c^2 + m_d^2)c^2 - s} = \left(2m_a^2 + m_c^2 + m_b^2\right)c^2 - \frac{2\sqrt{s}}{c}\mathcal{E}'_a$$

$$\Rightarrow \left(m_b^2 - m_a^2\right)c^2 - s = -\frac{2\sqrt{s}}{c}\mathcal{E}'_a, \quad \frac{\mathcal{E}'_a}{c} = \frac{s + (m_a^2 - m_b^2)c^2}{2\sqrt{s}}$$

$$\Rightarrow \frac{\mathcal{E}'_b}{c} = \frac{s - (m_a^2 - m_b^2)c^2}{2\sqrt{s}}$$

$\Rightarrow$ can find momentum $|\vec{p}_a'| = \sqrt{\frac{\mathcal{E}_a^2}{c^2} - m_a^2 c^2}$

$$|\vec{p}_a'|^2 = \left(\frac{s + (m_a^2 - m_b^2)c^2}{2\sqrt{s}}\right)^2 - m_a^2 c^2$$

$$= \frac{1}{4s}\left[s^2 + (m_a^2 - m_b^2)^2 c^4 + 2s(m_a^2 - m_b^2)c^2\right] - m_a^2 c^2$$

$$= \frac{1}{4}s + \frac{(m_a^2 - m_b^2)^2 c^4}{4s} - m_b^2 c^2$$

Clearly $|\vec{p}_b'| = |\vec{p}_a'|$. Can also find $|\vec{p}_a|$ (then $|\vec{p}_b| = 0$).

$$|\vec{p}_a'|^2 = \frac{1}{4s}\left[s^2 + (m_a^2 c^2)^2 + (m_b^2 c^2)^2 - 2(m_a^2 c^2)(m_b^2 c^2) - 2s(m_a^2 c^2) - 2s(m_b^2 c^2)\right] \equiv \frac{\lambda^2(s, m_a^2, m_b^2)}{4s}$$

where $\lambda(x,y,z) = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz$.

In the lab frame: $|p_a| = \sqrt{\frac{\mathcal{E}_a^2}{c^2} - m_a^2 c^2} = \left[\left(\frac{s - m_a^2 c^2 - m_b^2 c^2}{2 m_b c}\right)^2 - m_a^2 c^2\right]^{1/2} =$

$$= \frac{1}{2 m_b c}\left[\left(s - (m_a^2 + m_b^2)c^2\right)^2 - 4(m_b^2 c^2)(m_a^2 c^2)\right]^{1/2} =$$

$$= \frac{1}{2 m_b c}\left[s^2 + (m_a^2 c^2)^2 + (m_b^2 c^2)^2 - 2s(m_a^2 c^2) - 2s(m_b^2 c^2) - 2(m_a^2 c^2)(m_b^2 c^2)\right]^{1/2}$$

$$= \frac{1}{2 m_b c}\lambda^2(s, m_a^2, m_b^2).$$

So we found $(\vec{p}\,'_a)^2 = \dfrac{\lambda^2(s, m_a^2, m_b^2)}{4s} = (\vec{p}\,'_b)^2$

Since $s = \frac{1}{c^2}(\mathcal{E}'_a + \mathcal{E}'_b)^2 = \frac{1}{c^2}(\mathcal{E}'_c + \mathcal{E}'_d)^2$

and $\vec{p}\,'_a + \vec{p}\,'_b = 0 = \vec{p}\,'_c + \vec{p}\,'_d$

$$\Rightarrow (\vec{p}\,'_c)^2 = (\vec{p}\,'_d)^2 = \frac{\lambda^2(s, m_c^2, m_d^2)}{4s} \quad \text{too.}$$

To find angle between $\vec{p}\,'_a$ and $\vec{p}\,'_c$, use

$$t = (p_a - p_c)^\alpha (p_a - p_c)_\alpha = m_a^2 c^2 + m_c^2 c^2$$

$$- p_a^\alpha p_{c\alpha} - \underbrace{p_c^\alpha p_{a\alpha}}_{p_a^\alpha p_{c\alpha}} = m_a^2 c^2 + m_c^2 c^2 - 2\left(\frac{\mathcal{E}'_a}{c}\frac{\mathcal{E}'_c}{c} - \vec{p}\,'_a \cdot \vec{p}\,'_c\right)$$

$$\Rightarrow 2 p'_a p'_c \cos\theta'_{ac} = t + 2\frac{\mathcal{E}'_a \mathcal{E}'_c}{c^2} - m_a^2 c^2 - m_c^2 c^2$$

But $2\dfrac{\mathcal{E}'_a}{c}\dfrac{\mathcal{E}'_c}{c} = \dfrac{s + (m_a^2 - m_b^2)c^4}{2\sqrt{s}} \cdot \dfrac{s + (m_c^2 - m_d^2)c^4}{2\sqrt{s}} \cdot 2$

$$= \frac{1}{2s}\Big\{s^2 + s\left[(m_a^2 - m_b^2)^2 c^4 + (m_c^2 - m_d^2)^2 c^4\right] + (m_a^2 - m_b^2)^2 c^4 (m_c^2 - m_d^2)^2 c^4\Big\}$$

$$2 p'_a p'_c \cos\theta'_{ac} = t + s^2 + s\left[(m_a^2 - m_b^2)^2 c^4 + (m_c^2 - m_d^2) c^4\right.$$

$$2 p'_a p'_c \cos\theta'_{ac} = t + 2\frac{\mathcal{E}'_a \mathcal{E}'_c}{c^2} - m_a^2 c^2 - m_c^2 c^2$$

$$2\frac{\mathcal{E}'_a}{c}\frac{\mathcal{E}'_c}{c} = \frac{2}{4s}\left(s + (m_a^2 - m_b^2)c^2\right)\left(s + (m_c^2 - m_d^2)c^2\right)$$

$$= \frac{1}{2s}\left[s^2 + s(m_a^2 - m_b^2 + m_c^2 - m_d^2)c^2 + (m_a^2 - m_b^2)(m_c^2 - m_d^2)c^4\right.$$

$$2 p'_a p'_c \cos\theta'_{ac} = \frac{1}{2s}\left\{2st + s^2 + s(m_a^2 - m_b^2 + m_c^2 - m_d^2)c^2\right.$$

$$\left. - 2sc^2(m_a^2 + m_c^2) + (m_a^2 - m_b^2)(m_c^2 - m_d^2)c^4\right\}$$

$$= \frac{1}{2s}\left\{s^2 + s\left[2t - (m_a^2 + m_b^2 + m_c^2 + m_d^2)c^2\right] + (m_a^2 - m_b^2)(m_c^2 - m_d^2)\cdot c^4\right\}$$

But $2 p'_a p'_c = \frac{2}{\sqrt{16 s^2}}\lambda^{1/2}(s, m_a^2, m_b^2)\,\lambda^{1/2}(s, m_c^2, m_d^2)$

$$\Rightarrow \cos\theta'_{ac} = \frac{s^2 + s\left[2t - (m_a^2 + m_b^2 + m_c^2 + m_d^2)c^2\right] + (m_a^2 - m_b^2)(m_c^2 - m_d^2)c^4}{\lambda^{1/2}(s, m_a^2, m_b^2)\,\lambda^{1/2}(s, m_c^2, m_d^2)}$$